Dataset Collection for Urdu Named Entity Recognition

Decision Report: Web Scraping with LLM Annotation

1. Purpose

Our project builds an Urdu Named Entity Recognition (NER) system, but publicly available Urdu NER datasets are few, small and inconsistent. This report compares two ways to obtain training data and explains why we chose to scrape web pages and annotate them with Large Language Models (LLMs).

2. Options Considered

Option A: Scrape Urdu web pages and annotate the text with an LLM. **Option B:** Merge the existing Urdu NER datasets we explored and align their labels to one common definition.

Criterion	Option A: Web scraping + LLM annotation	Option B: Merge existing datasets
Data volume	Scalable; limited only by crawl size and API budget.	Fixed and small; Urdu NER corpora are scarce.
Label consistency	One schema and one prompt, applied uniformly.	Each dataset has its own tag set and guidelines; mapping loses detail.
Domain coverage	We choose domains (news, sports, politics, social).	Tied to the original domains, mostly news.
Annotation quality	Needs validation; LLMs can err on Urdu.	Human-annotated, but of varying quality.
Cost and effort	API cost plus a human-review budget.	Low cost, but heavy label-mapping work.

3. Decision

We selected Option A. The main limitation of Option B is that the existing corpora are small and cover few domains. Merging them also requires mapping different tag sets (for example, differing treatment of organisations, locations or miscellaneous entities), which causes information loss and residual inconsistency. Option A gives us control over size, domain and the label schema, and an LLM can annotate a large volume of text far faster than manual labelling. Its main risk, annotation noise, is handled with the validation steps below. The existing datasets are not discarded: we keep them as a gold-standard benchmark for evaluating the quality of our new data.

4. Proposed Pipeline, Techniques and Tools

- **Schema definition:** Fix one tag set (PER, LOC, ORG, plus optional DATE and MISC) with written guidelines and examples, in BIO or IOB2 format.
- **Web scraping:** Collect Urdu news, sports, business and social content using Scrapy, BeautifulSoup, Requests and Selenium or Playwright for dynamic pages, respecting robots.txt and site terms.
- **Cleaning and preprocessing:** Remove HTML and boilerplate with trafilatura; normalise Unicode and Urdu characters (diacritics, Arabic and Persian letter variants); detect language using fastText or langdetect; deduplicate with MinHash; split sentences and tokenise using UrduHack or spaCy.
- **LLM annotation:** Use a strong multilingual LLM (such as Claude, GPT-4 class or Gemini) through its API with few-shot, schema-constrained prompts and structured JSON output. Run each sentence through the model more than once and keep entities by majority vote (self-consistency) to reduce noise.
- **Quality control:** Convert LLM output to token-level BIO tags and check spans align with tokens. Have native Urdu speakers review a random sample (about 5 to 10 percent) in Label Studio or Doccano, and measure agreement with Cohen's kappa and span-level F1. Revise the prompt and re-annotate if quality is low.
- **Model training and evaluation:** Fine-tune multilingual transformers (XLM-RoBERTa, mBERT, MuRIL) with Hugging Face Transformers and evaluate with seqeval (precision, recall, F1) on the human-reviewed set and on the existing datasets as an independent benchmark.

5. Risks and Mitigation

- **LLM errors on Urdu names and rare entities:** mitigated by few-shot examples, voting across runs and human spot checks.
- **Domain bias and duplicates:** mitigated by crawling several sources and deduplicating.
- **Legal and ethical concerns:** mitigated by using public pages, honouring robots.txt, and recording data sources and licences.
- **API cost:** mitigated by annotating in batches and starting with a pilot of 1,000 to 2,000 sentences.